SEVEN EQUIVALENT CHARACTERISATIONS OF GEOMETRIC REDUCEDNESS IN POSITIVE CHARACTERISTIC

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-varieties/retained-preamble.tex), a verbatim excerpt of the native Xy-pic setup, of the native theorem declarations with mathematical macros, and of the project's own no-op definitions of the reference display environments; no macro was invented or redefined.

Editorial note. All four referenced results live in earlier chapters, so their locators point at the verbatim lines of the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt) that carry the canonical tags.

Editorial note. Overlap pre-check: this label occurs in no other staged chapter (run with agent-authoring/stk/card_plan.py varieties before opening).

Allowed standard prerequisites (editorial).

- Geometrically reduced algebra over a field, and reducedness transferred to fibres. Algebra, Definition labelled algebra-definition-geometrically-reduced (canonical Stacks tag 030S); named in the counted statement at line 431. Supplies the algebraic notion used in condition (7); the passage between the ring conditions is the counted content.
- Reducedness of a ring and of its base change, and integrality of the structure sheaf. Properties, Section labelled properties-section-integral (canonical Stacks tag 01OJ); invoked at proof lines 444-445. Supplies that reducedness of Spec(O(U) tensor_k k') is reducedness of the ring, and the gluing description of the base change.
- Fibre products of schemes are obtained by gluing the spectra of the tensor products over an affine cover. Schemes, Section labelled schemes-section-fibre-products (canonical Stacks tag 01JO); invoked at proof lines 450-452. Supplies the description of X_{k'} by gluing spectra of O_X(U) tensor_k k', used in the implication (7) => (1).
- It suffices to test geometric reducedness on finite purely inseparable extensions. Algebra, Lemma labelled algebra-lemma-geometrically-reduced-finite-purely-inseparable-extension (canonical Stacks tag 030V); invoked at proof lines 457-459. Supplies the equivalence of (1) with conditions (3)-(6) by applying it to the rings O_X(U).

Source reference. Original definition 030S: sources/stacks-chow/tags.txt, source lines 3959–3959. Cross-chapter pointer named in the counted statement at line 431 for the ring-theoretic notion; the equivalence proof is retained in the counted span.

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Source reference. Original section 01OJ: sources/stacks-chow/tags.txt, source lines 2196–2196. Cross-chapter pointer invoked once at proof lines 444-445 for the reading of reducedness of the spectrum as reducedness of the ring.

Source reference. Original section 01JO: sources/stacks-chow/tags.txt, source lines 2021–2021. Cross-chapter pointer invoked once at proof lines 450-452 for the gluing description of the base change X_{k'}.

Source reference. Original lemma 030V: sources/stacks-chow/tags.txt, source lines 3962–3962. Cross-chapter standard prerequisite invoked once at proof lines 457-459 to attach conditions (3)-(6); the cycle (1)-(2)-(7) is retained in the counted span.

Lemma 0.1. *Let k be a field of characteristic $p > 0$. Let X be a scheme over k. The following are equivalent*

(1) *X is geometrically reduced,*
(2) *$X_{k'}$ is reduced for every field extension k'/k,*
(3) *$X_{k'}$ is reduced for every finite purely inseparable field extension k'/k,*
(4) *$X_{k^{1/p}}$ is reduced,*
(5) *$X_{k^{perf}}$ is reduced,*
(6) *$X_{\bar{k}}$ is reduced,*
(7) *for every affine open $U \subset X$ the ring $\mathcal{O}_X(U)$ is geometrically reduced (see Algebra, Definition 030S).*

Proof. Assume (1). Then for every field extension k'/k and every point $x' \in X_{k'}$ the local ring of $X_{k'}$ at x' is reduced. In other words $X_{k'}$ is reduced. Hence (2).

Assume (2). Let $U \subset X$ be an affine open. Then for every field extension k'/k the scheme $X_{k'}$ is reduced, hence $U_{k'} = \mathrm{Spec}(\mathcal{O}(U) \otimes_k k')$ is reduced, hence $\mathcal{O}(U) \otimes_k k'$ is reduced (see Properties, Section 01OJ). In other words $\mathcal{O}(U)$ is geometrically reduced, so (7) holds.

Assume (7). For any field extension k'/k the base change $X_{k'}$ is gotten by gluing the spectra of the rings $\mathcal{O}_X(U) \otimes_k k'$ where U is affine open in X (see Schemes, Section 01JO). Hence $X_{k'}$ is reduced. So (1) holds.

This proves that (1), (2), and (7) are equivalent. These are equivalent to (3), (4), (5), and (6) because we can apply Algebra, Lemma 030V to $\mathcal{O}_X(U)$ for $U \subset X$ affine open. □